

Omni-Channel Retail Inventory Analytics Report

Report prepared by team G-1

Introduction:

This project focuses on integrating and analyzing sales, inventory, and customer data to provide a unified analytical view that supports better business decision-making.

Problem Statement: Retail businesses face several challenges:

- Lack of unified visibility across sales channels
- Poor synchronization between sales and inventory data
- Difficulty in tracking customer behavior and retention
- Inefficient inventory management leading to stockouts or overstock

Objective: The main objectives of this project is:

- Clean and preprocess raw datasets
- Ensure data consistency and integrity
- Build a scalable data model
- Develop an interactive dashboard
- Generate actionable business insights

Executive Summary

This omnichannel retail operation demonstrates a balanced but challenging business. Total revenue reached approximately **₹382.6 million** across ~50,000 orders.

The three channels (App, Online, Store) show remarkably even performance, each contributing roughly one-third of revenue and orders.

Key Highlights:

- **Revenue distribution:** Nearly equal across App (₹128M), Online (₹127M), and Store (₹128M).
- **Order Status:** Roughly 25% each for Delivered, Shipped, Returned, and Cancelled — indicating significant leakage in fulfillment (high return/cancellation rates ~25% each).
- **Top Performers:** Dell, Noise, Ikea, and Samsung products lead sales.

- **Inventory Health:** Low turnover ratios (highest ~0.46) and high stock-to-sales ratios signal overstocking and capital tie-up.
- **Customer Base:** ~5,000 customers with average ~5 orders each; strong loyalty in legacy segment.

Overall Assessment: Solid topline with even channel mix, but profitability is likely eroded by high returns/cancellations and inefficient inventory management.

1. Sales Performance

Core Metrics:

- **Total Revenue:** ₹382,613,395
- **Total Orders:** ~50,000
- **Average Order Value (AOV):** ~₹7,600 (with high variation; SD ~₹4,900)
- **Units Sold (Top Products):** Dell Item_452 (621), Noise Item_438 (589), H&M Item_637 (573), etc.

Monthly Revenue Trend (notable peaks/lows):

- Strongest: Jan 2023 (~₹15.7M+ net), Dec 2024, Sep 2023
- Weakest: Apr 2023, Jun 2023, Sep/Oct 2024

Channel Performance:

- **App:** Strongest digital channel (~₹128M)
- **Online:** Very close to App
- **Store:** Competitive with digital despite physical constraints

Digital vs Physical:

- Digital (App + Online): ~66% of orders, ~₹255M revenue
- Physical (Store): ~34% of orders, ~₹128M revenue

Return & Cancellation Rates: ~25% each across channels — a critical area for improvement.

2. Product Performance

Top Selling Products (Delivered/Shipped):

1. Ikea Item_519 (Home) — 323 units
2. Samsung Item_630 (Fashion) — 308 units
3. Ikea Item_418 (Home) — 301 units

Highest Revenue Products (Delivered/Shipped):

- Dell Item_452 (Sports) — ~₹565K
- Noise Item_330 (Fashion)
- Zara Item_295 (Fashion)

Problem Products:

- High cancellation: Dell Item_452, Puma Item_745
- High returns: Boat Item_701, Samsung Item_193
- Lowest revenue: H&M Item_425, Decathlon Item_556

Category Leaders:

- Electronics: Noise
- Fashion: Samsung
- Home: Ikea
- Sports: Dell

3. Customer Analysis

- **Total Customers:** ~5,000
- **Repeat Customers:** Significant portion (many with >1 order)
- **Top Customer:** Leo Kapoor (Delhi) — highest revenue; Mia Rossi (Mumbai) — highest frequency (15 orders)
- **City Performance:** Chennai leads in customer count and revenue, followed by others with relatively even distribution.

Cohort Insights:

- Legacy customers (pre-2023) show strong loyalty and increasing frequency.
- 2024 new customers: Sharp drop in numbers and engagement.

Average Lifetime Value: ~₹38K per customer (high variance).

4. Inventory & Supply Chain Analysis

Current Stock: Significant volume across 10 warehouses.

Key Issues:

- **Low Inventory Turnover:** Highest ratio only 0.46 (Ikea Item_54). Many products below 0.25.
- **Stock-to-Sales Ratio:** Many products heavily overstocked (ratios up to 133+ months of supply).
- **Movements:** Balanced restocks vs sales, but poor velocity overall.

Recommendation: Major rationalization needed — reduce safety stock, improve demand forecasting, and implement better replenishment logic.

5. Store & Warehouse Performance

- **Online channel** dominates overall revenue.
- Top physical stores (e.g., STORE024 in Hyderabad, STORE014 in Chennai) perform well but lag far behind online.
- Warehouses show varying capacities (12K–49K units); utilization and allocation need review.

6. Strategic Insights & Recommendations

Strengths:

- Healthy, balanced omnichannel presence.
- Strong brand mix (Dell, Ikea, Samsung, Noise).
- Loyal legacy customer base.
- Good geographic spread across major Indian cities.

Weaknesses:

- High return (25%) and cancellation (25%) rates — hurting net revenue and customer experience.
- Poor inventory turnover and overstocking — tying up capital.
- Declining new customer acquisition in 2024.
- Some products consistently underperform or drive returns.

Recommendations:

1. **Return & Cancellation Reduction:**
 - Root cause analysis on top problematic products.
 - Improve product descriptions, sizing guides, and quality checks.
2. **Inventory Optimization:**
 - Adopt ABC analysis + EOQ models.
 - Reduce stock on slow-moving items.
 - Improve sales-to-stock forecasting using historical + seasonal trends.
3. **Customer Acquisition & Retention:**
 - Targeted campaigns for new customer onboarding in 2025.
 - Loyalty program enhancements for high-frequency buyers.
 - Personalized marketing by city and segment.
4. **Channel Synergies:**
 - Leverage strong digital performance for "Buy Online, Pick Up in Store" (BOPIS) to boost physical stores.

- Cross-channel inventory visibility.
- 5. **Data & Analytics:**
 - Implement real-time dashboards
 - Monitor cohort retention and CLV monthly.
 - Add margin/profitability tracking.

Conclusion:

The business has a solid foundation with balanced channel performance and popular product lines. However, **net profitability is likely constrained** by high reverse logistics costs and inefficient inventory.

Focusing on **reducing returns/cancellations by 10–15 percentage points** and **improving inventory turnover** could unlock significant margin improvement and free up working capital.